

Integrate a pre-engineered microfracture lattice into the drainage floor surface, creating a passive network of channels that open under hydraulic pressure to compensate for viscosity-induced resistance.

Executive Technology Brief

Domain: fluidics_hydraulic

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'hydraulic performance deviation' concerning drainage floor. User-submitted engineering problem (session input): "Why does P-07's drainage floor deviate from the measured water viscosity basis? (reality discrepancy semantics)". [fra_rail_accidents] retrieval status GRAMMAR_MISMATCH (query does not match source grammar: YYYY-MM-DD date floor (\$where date >= X) — engine routing error, not a provider failure (not absence, Art. XXV)) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] 10 records retrieved live (2026-2026). Illustrative records: 'Evaluating the Analytical Capabilities of RAM In-situ Diagnostics'; 'Resonant Acoustic Mixing of Dense Pastes: Role of Adhesion and Viscosity'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: Porefracture structure provides passive hydraulic pathways that modulate flow resistance.

Architecture: PROPOSED architecture (no physical system exists): flow path / lumen architecture; pressure regulation element; sensing / feedback element (if closed-loop); termination interfaces (patient / reservoir) — realizing: Integrate a pre-engineered microfracture lattice into the drainage floor surface, creating a passive network of channels that open under hydraulic pressure to compensate for viscosity-induced resistance.

Governing Model: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

FLUID-001: Hagen-Poiseuille (laminar pipe flow). Expression: $Q = (\pi * r^4 * \Delta P) / (8 * \mu * L)$. Applicability to this invention: APPLICABLE; holds under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow. Assumptions: CSF approximated as Newtonian (valid at shunt flow rates); no wall compliance effects. Engaged variables: ΔP , μ .

FLUID-002: Reynolds number. Expression: $Re = (\rho * v * D) / \mu$. Applicability to this invention: APPLICABLE; holds under: circular section; single-phase flow. Assumptions: transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb. Engaged variables: D , μ .

FLUID-003: Orifice flow (sharp-edged restriction). Expression: $Q = C_d * A * \sqrt{2 * \Delta P / \rho}$. Applicability to this invention: APPLICABLE; holds under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured. Assumptions: $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement. Engaged variables: ΔP .

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.